Weekly accident and emergency (A&E) demand and escalation briefing: design case study

Patrick Gilliland, 27 October 2026

Every figure in this case study comes from the spot-check notebook, `ae_escalation_spot_check.ipynb`, or from a source in the reference list. The notebook's figures are in-sample: the design was tuned on the same 2015 to 2026 data that scores it.

Terms. A week is busy when its attendances reach 1.05 times its simulated capacity, and extra patients are attendances above simulated capacity in busy weeks. An outlook is one hospital's four weeks ahead at one origin; a hospital-week counted once is one hospital's one week, however many outlooks covered it. The drift alarm fires when a series' actual attendances fall outside the range its forecast expected for two released weeks running. Once it fires, the drift latch stays set, like an alarm that stays on until someone resets it. While it is set, the forecasts are marked untrusted. In the design only a person can reset it; in this build each reset is simulated at its review prompt. A latch after an upward breach is a surge, and its level comes from the latest released week; after a downward breach the level is held at what the flag gave the week before the alarm. The steady years are 2018 from July, 2019 and 2023 to 2026, and the mean steady year is the mean of 2019, 2023, 2024 and 2025. Each figure caption gives the figure's number in the notebook, which its file name in `figures/` carries.

1. Overview

Industry: healthcare operations, unplanned attendances at Scotland's emergency departments, for the duty managers who staff the front door.

Status: additional information only. At this stage the system is a pilot build that gives the duty manager additional information beside their existing process, by choice. It replaces no current process and sets no staffing on its own. Becoming a primary operational tool needs shadow mode, a pilot with managers who did not build it and tuning over time, which is achievable but not yet done (section 14).

The problem. Emergency department rotas are set weeks ahead. In England the rota code expects resident doctors in training to get duty rosters at least 6 weeks before a new rotation starts (NHS England, 2025). So when a week turns out unusually busy, the front door is staffed for the usual week. Scottish guidance says emergency department crowding "is linked to increased morbidity and mortality" (Scottish Government, 2016, Background section). A forecast one to four weeks ahead can still inform short-notice changes inside a rota already issued: extra shifts, bank staff or holding leave.

The users. A health board's duty managers, who staff the front door, and its unscheduled-care lead, who owns the thresholds and approves each latch reset.

The demand level, in one sentence. The demand level shows demand unusually high for the time of year; it is not pressure, need or priority.

What is real and what is simulated.

Input	Real or simulated
Weekly unplanned attendances per hospital, and their national sum, from Public Health Scotland (PHS, 2026a)	Real
Attendances within 4 hours and waits over 12 hours, used only for the month chart and the pressure match	Real
Scottish bank holidays, GOV.UK's bank-holidays JSON	Real
The seasonal capacity the level is measured against	Simulated: a demand percentile, not beds or staff

Input	Real or simulated
The bands, 1.05, 1.10 and 1.15 times capacity	My design, illustrative
The staffing actions by level, their owners and reply deadlines	My design, illustrative
Staff acknowledgements of notifications	Simulated and seeded
Latch resets	Simulated: recorded at each review prompt
The language model's replies	Recorded live, replayed exactly in the notebook

The value chain

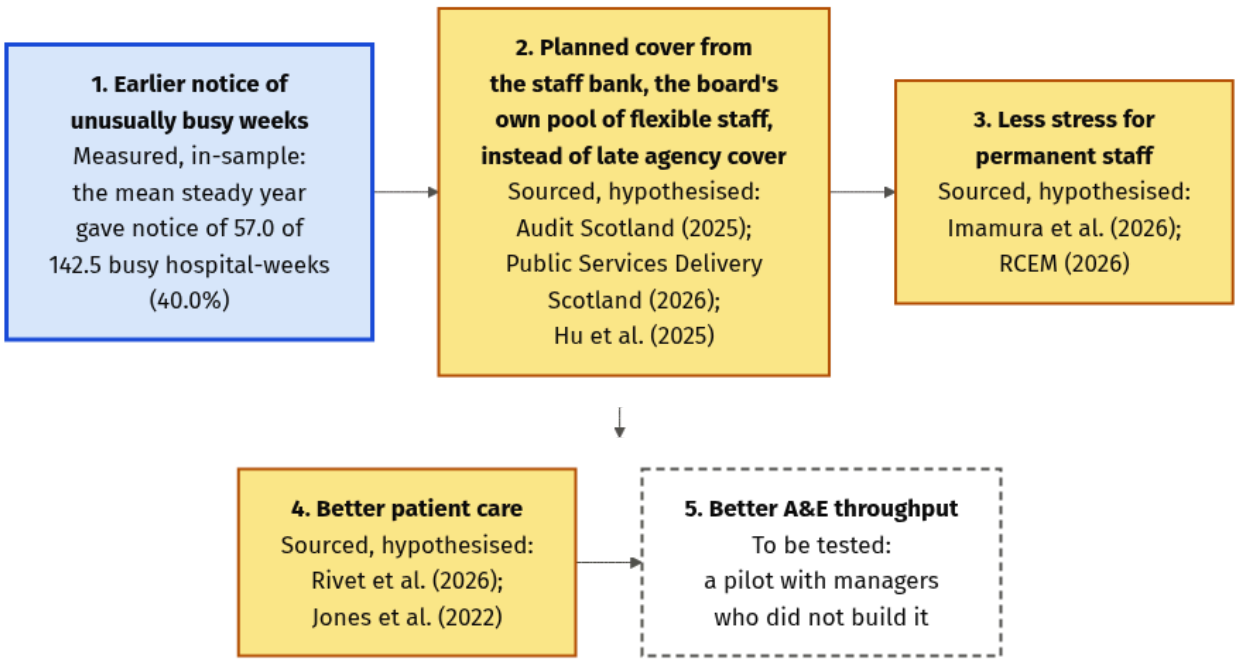


Figure 1 (notebook Figure 2). The value chain: measured, sourced, to be tested. Source: `diagram/value_chain.mmd`.

#	Link	Label	Evidence
1	Earlier notice of unusually busy weeks	Measured, in-sample	The mean steady year gave notice of 57.0 of 142.5 busy hospital-weeks (40.0%). Over the steady years combined, 280 of 713 busy hospital-weeks had notice, carrying 19,806 of 46,243 extra patients; 197 had it from the week's own flag
2	Planned cover from the staff bank, the board's own pool of flexible staff, instead of late agency cover	Sourced, hypothesised	Audit Scotland (2025, p. 15), 2024/25: £256m on agency staff of all kinds (£57m nursing, £108m medical) and, separately, £353m on the nursing bank. Public Services Delivery Scotland (2026), 2025/26: agency nursing and midwifery £34.5m (down 39.2%), nursing and midwifery bank £391.0m (up 10.6%) on hours up 0.1%. Hu et al. (2025): one US emergency department's forecast-driven nurse staffing cut hourly staffing costs by $162.04 with no loss of throughput, before and after, with no control group
3	Less stress for permanent staff	Sourced, hypothesised	Imamura et al. (2026): a few studies of burnout and turnover suggest potential benefits of adequate staffing. Royal College of Emergency Medicine (RCEM, 2026): in its December 2025 survey one Scottish emergency department clinical lead reported decision fatigue and burnout among staff, and RCEM links crowding to burnout; the survey does not give its response count

#	Link	Label	Evidence
4	Better patient care	Sourced, hypothesised	Rivet et al. (2026): more nurses per patient associated with lower emergency department mortality (adjusted odds ratio 0.68, 0.49 to 0.94), observational. Jones et al. (2022): delays to admission associated with 30-day mortality. RCEM (2026) estimates 818 deaths in Scotland in 2025 associated with long waits: an estimate, not a count
5	Better A&E throughput	To be tested	The pilot's goal (section 14)

Link 1 is measured, in-sample. Links 2 to 4 are benefits the sources support, and link 5 is a hypothesis a pilot would test. None is an effect this system has shown.

The scale. In 2024/25 Scotland spent £256m on agency staff of all kinds and, separately, £353m on the nursing bank, about £0.6bn together (Audit Scotland, 2025). Scotland's A&E sites, minor injury units included, saw 137,703 attendances a month in 2025, about 1.65m in the year (PHS, 2026b). The weekly data holds 1,405,613 attendances at the 30 emergency departments for weeks ending in 2025, my sum in the notebook (PHS, 2026a). At that scale, a small share of that spend is a large sum.

The caution. The money saving would be mainly the agency premium, agency hours moved to planned bank hours, not the whole supplementary spend, and the agency figure covers every specialty, not only A&E. The wider value lies in links 3 and 4 of the value chain, less stress for permanent staff and better patient care, which are sourced and hypothesised, not measured here. I quote figures on lives or harm only to show what is at stake; this system has not shown any effect on them.

The wider system. The data holds attendances, which run highest in May and June, while the share seen within 4 hours is lowest in December and January, and waits over 12 hours peak in January (Figure 2). It holds no beds, so uses for beds and social care need admissions and occupancy data (section 14).

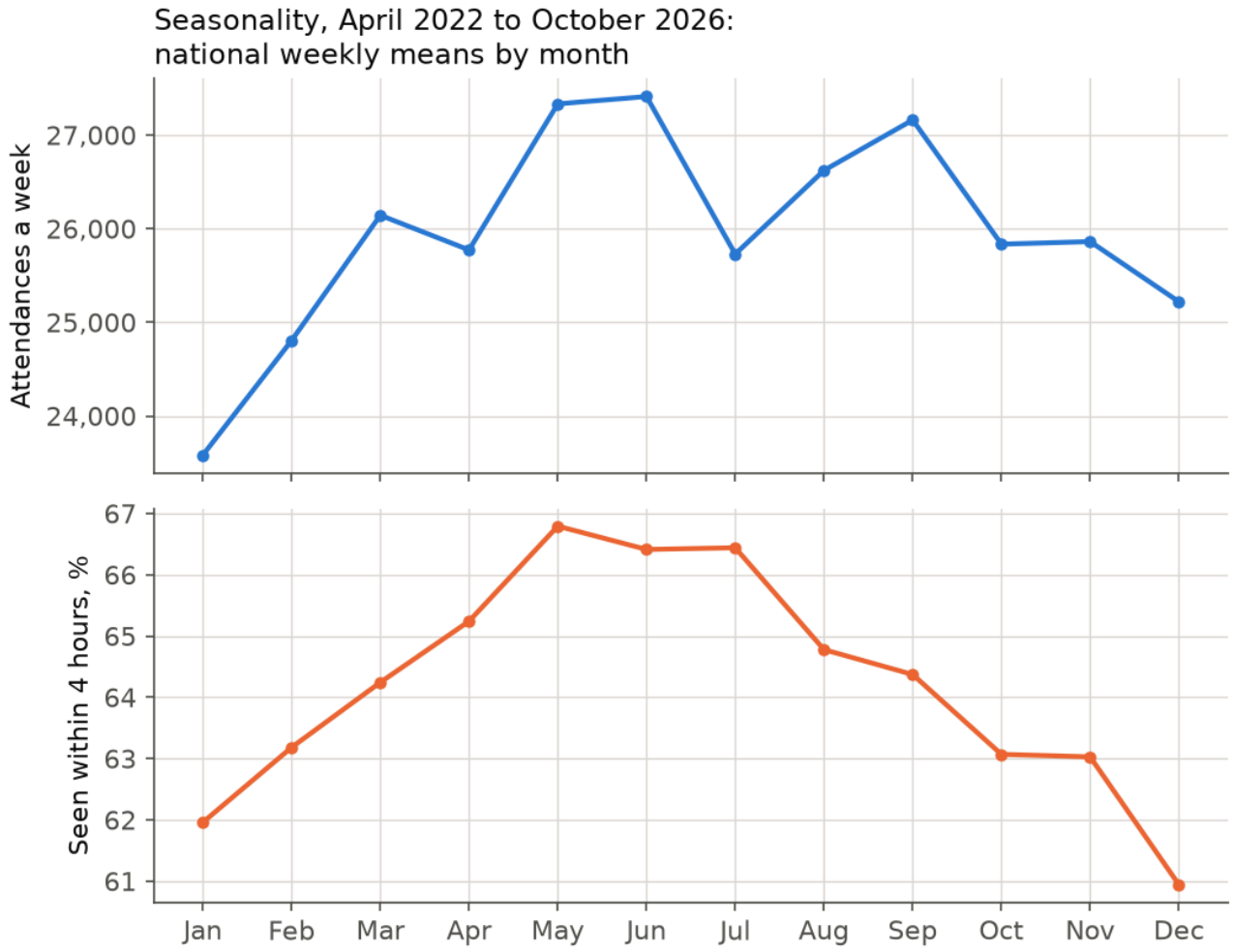


Figure 2 (notebook Figure 4). Monthly attendances against 4-hour performance. Seasonality only: PHS says these weekly figures "should not be used to monitor against the 4-hour A&E access standard" (PHS, 2026a, data set description).

2. Architecture

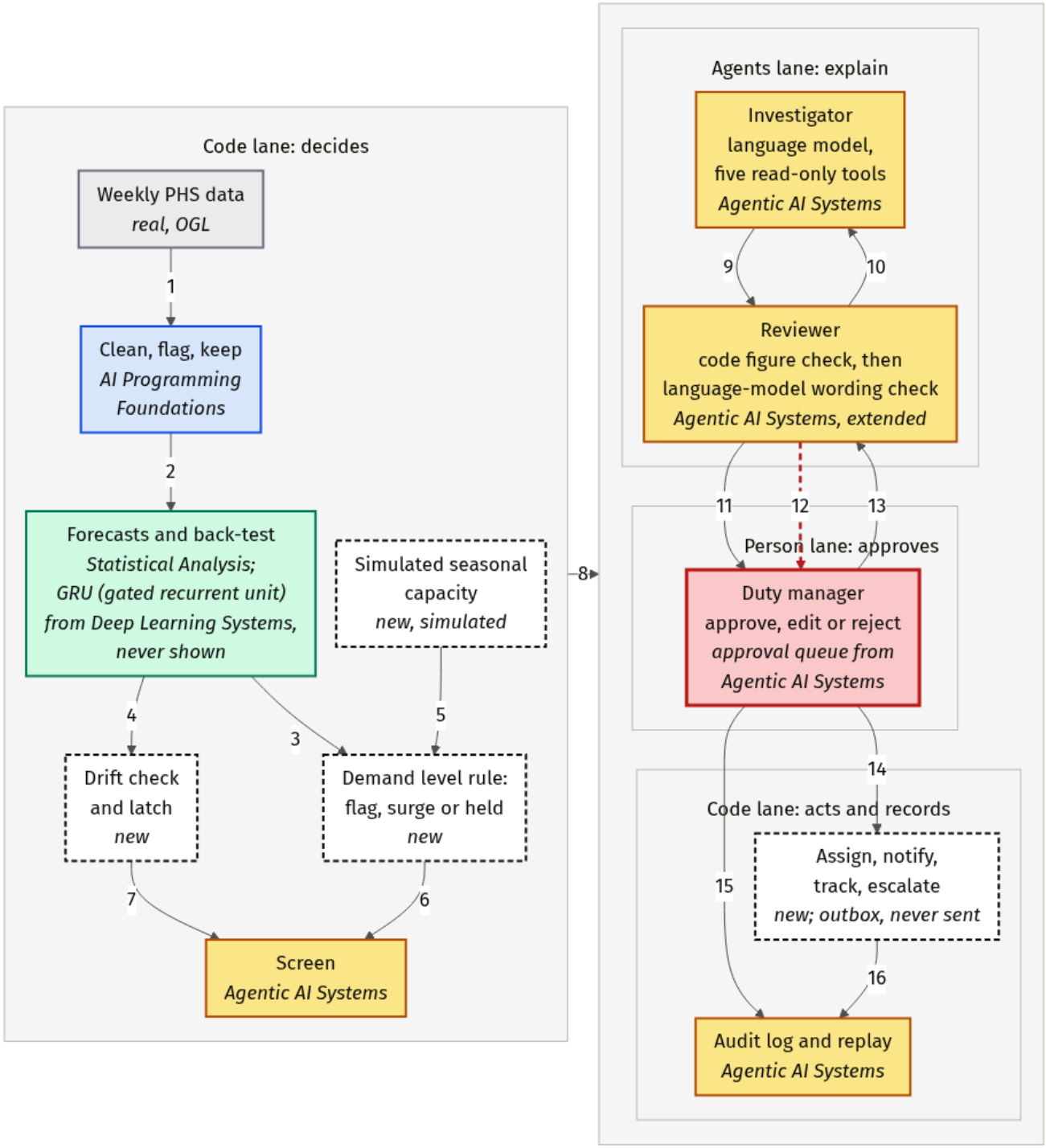


Figure 3 (notebook Figure 1). Code decides, the agents explain, a person approves. Source: `diagram/architecture.mmd`.

The design has three lanes. **Code decides**: it loads the data, forecasts, sets the demand level, checks for drift, screens which hospitals reach the agents, and then acts and records. **The agents explain**: an Investigator drafts each latched week's demand brief from read-only tools, and a Reviewer checks it; in a flag week code fills the template brief. **A person approves**: the duty manager approves, edits or rejects each brief. The screen is code, so it sits in the code lane. Figure 3 draws the code lane in two parts, before and after the person decides: the first on the left, and the second last on the right, below the agents and the person.

Each box names the prior capstone project it comes from, or says it is new in this project:

- **Weekly PHS data**: real, under the Open Government Licence (OGL).
- **Clean, flag, keep**: AI Programming Foundations.
- **Forecasts and back-test**: Conduct a Statistical Analysis Using Python ("Statistical Analysis" below), with a comparator from Deep Learning Systems, a recurrent network, a GRU (gated recurrent unit), never shown to a manager.
- **Simulated seasonal capacity, the demand level rule, the drift check and latch, and the coordinator**: new in this project.
- **The screen, the Investigator, the Reviewer, the approval queue, and the audit log and replay**: Agentic AI Systems, the Reviewer extended with a wording check.

Each arrow carries only its number, which is its row in the interaction table in section 3.

3. Data flow and orchestration

The weekly flow, in nine steps.

1. **Weekly load.** Code reads the PHS file, checks its SHA-256, cleans it and flags anomalies, keeping every row. It uses only the weeks released by the forecast origin.
2. **Forecast.** Seasonal naive, the 4-week moving average, a linear model and SA-MA, the seasonally adjusted moving average, which is the 4-week moving average with a seasonal index, forecast the four week-endings after the release, each with an 80% and a 95% range. One primary model drives the flag. It starts as linear and is re-selected among the four at each latch reset. A GRU runs beside them as a comparator.
3. **Flag and drift.** From the primary's last 104 released errors, code works out each week's chance of reaching 1.05 times its simulated capacity, and flags an outlook when one week's chance is at least 50%. The drift alarm fires when actuals fall outside the drift interval two released weeks running, and sets the drift latch.
4. **Screen.** Code sends on only a hospital that is flagged, or latched, which covers a surge or held level of 2 or more. Nothing else reaches the agents.
5. **Investigator.** It drafts the demand brief through five read-only tools: forecast, history, back-test error, track record and calendar. Code writes every figure with a label saying what it is, and the level is always the rule's.
6. **Reviewer.** Code checks every figure against its tool result and the level against the rule. A language model then checks the wording: no claim of cause without a tool basis, no figure described as something its label says it is not, and nothing that reads as an instruction to a clinician.
7. **Approval.** The duty manager approves, edits or rejects, with initials, a time and the six approval criteria.
8. **Coordinate.** Code sets the staffing actions for the level, assigns each an owner and a reply deadline, tracks it and escalates it. The language model drafts only each notification's wording.
9. **Audit.** Every tool call, draft, check and decision is logged with the backend that wrote it, and every model call is recorded, so the notebook replays the run exactly. Each recording file's SHA-256 is in `recordings_manifest.tsv`, so a changed reply stops the replay as a changed request does.

The Tuesday release and the four release horizons. Each week is published on the Tuesday, 9 days after its week ends. A forecast origin is the latest released week, named by its week ending (w/e). The four release horizons are the four week-endings after the release. For the origin w/e 22 Mar 2020, released on Tue 31 Mar 2020, they are the weeks ending 5, 12, 19 and 26 Apr 2020, data horizons 2 to 5.

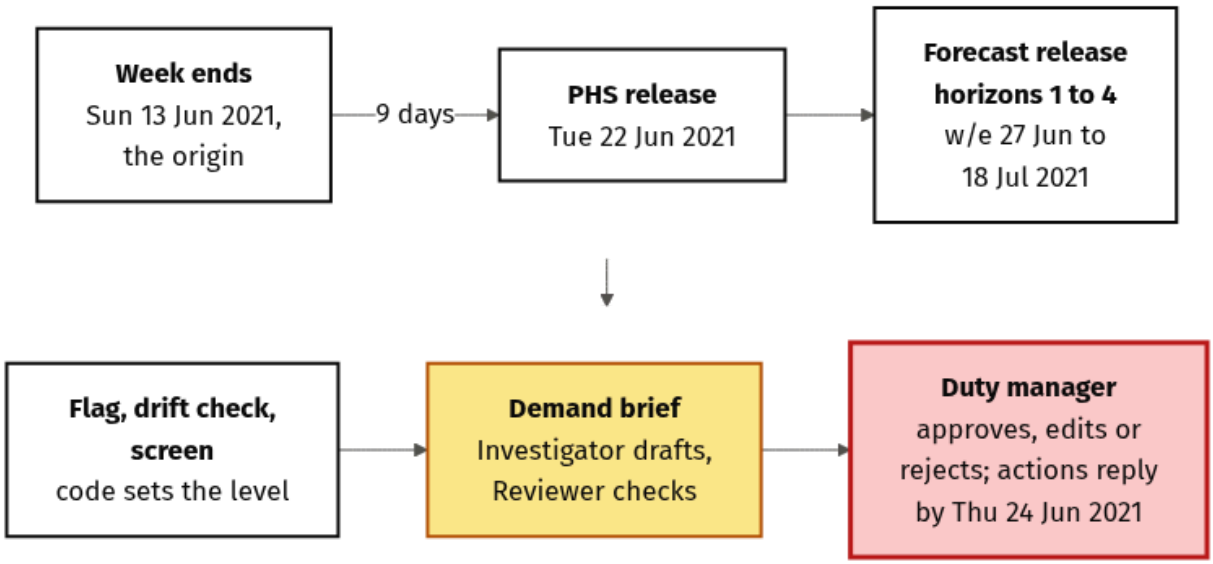


Figure 4 (notebook Figure 3). One week: Tuesday release to the duty manager's decision. Source: `diagram/weekly_cycle.mmd`. The dates illustrate the release timing for one replay week, Ninewells Hospital's demand week ending 13 Jun 2021; that week's brief was rejected.

The approval routes.

- A verified brief reaches the duty manager, who approves, edits or rejects it.
- A brief that fails a check goes back to the Investigator once, with every problem named. A brief that fails again reaches the manager marked "not verified, do not act", and can never be approved.
- An edit is checked again: the code figure check, the screen for the words the Medicines and Healthcare products Regulatory Agency (MHRA) lists as indicative of a medical device, and the Reviewer's wording check. An edit that fails comes back to the manager marked "not verified, do not act", never to the Investigator.
- A rejection is logged, and nothing is sent.

The coordinator's state machine. Level 1 sets no staffing action. At level 2, code sets two actions: check whether extra shifts or bank staff are needed, owned by the duty manager, and hold new leave requests, owned by the rota coordinator. Level 3 adds a request to the staff bank, and level 4 tells the unscheduled-care lead. Each action is notified; if its reply deadline passes without an acknowledgement, it is escalated to the role above, and a later acknowledgement is recorded as late. A notification that fails its figure check twice is held, and the manager sees it with its failing figures marked. The action's text is code's; the language model drafts the notification's wording, which states only figures the approved brief states.

The interaction table, one row for each numbered arrow in Figure 3. Each arrow carries only its number, and the table says what passes. Arrow 8 joins the left column to the right one; it runs from the screen to the Investigator.

#	From	To	What passes
1	Weekly PHS data	Clean, flag, keep	The weeks released by the origin, after the SHA-256 check
2	Clean, flag, keep	Forecast	The clean series per hospital and nationally, with its flags kept
3	Forecast	Demand level rule	The primary model's forecasts for the four release horizons, and its released errors, which give the flag's chance
4	Forecast	Drift check	The primary model's one-week-ahead forecast

#	From	To	What passes
5	Simulated seasonal capacity	Demand level rule	The capacity for each target week
6	Demand level rule	Screen	Hospitals flagged, or at a surge or held level of 2 or above, with the level's source
7	Drift check	Screen	The alarm, which sets the drift latch
8	Screen	Investigator	Each screened hospital and week; nothing else reaches the agents
9	Investigator	Reviewer	The draft demand brief, with each cited figure's tool-call id
10	Reviewer	Investigator	A failed brief, once, with the failing figures or wording named
11	Reviewer	Manager	A verified brief
12	Reviewer	Manager	A brief that failed twice, or an edit that failed, marked "not verified, do not act", failing figures marked
13	Manager	Reviewer	An edited brief, which the checks run again
14	Manager	Assign, notify, track, escalate	An approved, verified brief and the level's staffing actions
15	Manager	Audit log and replay	A rejection, logged; nothing is sent
16	Assign, notify, track, escalate	Audit log and replay	The actions, owners, outbox notifications and acknowledgements

Who writes a brief's explanation. In a new live run, `live briefs`, a flag week's brief is the template brief at version 2, filled by code with no model call, and a drift or surge week's, which is latched, is the Investigator's. This is the rule set before the template comparison's ratings, which the demand-week result triggered (section 13). The designed failure still goes to the Investigator. Each label records which way it ran, so the recorded runs, where the Investigator wrote every brief, replay as they ran. A template brief goes to the same approval queue: code checks every figure it prints against the tool call it came from, the duty manager approves, edits or rejects it, an edit is checked again, and only an approved brief sets actions (`examples/template_brief_ninewells.md`).

4. Assumptions

Items 1 to 7 are assumed, each with where it comes from; item 8 says how the spot check was run.

1. **Simulated capacity.** Assumed, my design: for each target week, the 90th percentile of the hospital's weekly unplanned attendances in the same 13-week window, centred on that week, in each of the previous three years, released weeks only. The open data holds no beds or staffing.
2. **The bands, 1.05, 1.10 and 1.15 times capacity.** Assumed, my design, on the structure of England's Operational Pressures Escalation Levels (OPEL) framework, where a score sets the level and people decide the actions, not its scale (NHS England, 2024).
3. **The flag's setting, a week's chance of at least 50%.** Assumed, chosen in the design probes by a rule stated before their results, on the same data: in-sample.
4. **The staffing actions by level and their owners.** Assumed, on the structure of OPEL's actions by level, each with an owner, not its actions (NHS England, 2024).
5. **The reply deadlines.** Assumed: 48 hours at level 2, 24 at level 3 and 12 at level 4, from 10:00 UK time on release day.
6. **The acknowledgements.** Assumed, simulated and seeded: three in four actions are acknowledged, after 2 to 72 hours.
7. **Latch resets.** Assumed: recorded at each review prompt after 8 clear weeks. In use the unscheduled-care lead approves each one.
8. **My three roles in the spot check.** I play the approver, the unscheduled-care lead and the analytics lead; the latch resets and re-selections in the replay are recorded by code at each

prompt. The clinical safety reviewer's sign-off is not exercised, because there is no live use.

5. Tradeoffs

Technical: a simple primary against the GRU. The primary is one of four simple models, re-selected at each latch reset. The GRU runs as a comparator only. Outside each series' drift episodes, the window that G1, forecasts a manager can plan with, scores, no release horizon of the five replay series had a paired interval favouring the GRU, and 16 of the 20 favoured the primary. Outside March 2020 to March 2022, the data test's window, one did: the Queen Elizabeth University Hospital at release horizon 4, by 0.56 points (0.06 to 1.01). As I stated before the run, the GRU stays a comparator for operational reasons. The gain from the simple primary is a forecast a manager can check in one sentence, with no yearly refit and no seeds to average. The cost is a forecast that cannot learn patterns the four simple models miss.

Data: simulated capacity against no capacity at all. With no beds or staffing in the open data, the level needs a line to measure demand against, so I simulate one from past demand. The gain is a level that can be tested on real attendances; the cost is a level that is illustrative, not a statement about beds. The sensitivity table shows what the setting changes: in the steady years, 22.2% of unlatched outlooks are flagged at the 80th percentile, 9.6% at the 90th and 4.8% at the 95th.

Operational: rarity against recall. The flag fires on 9.5% of the steady-year outlooks whose weeks are known, and 53.8% of those flags are borne out, against 21.2% of outlooks without one. The sensitivity table above counts every unlatched outlook, so its rate at the 90th percentile is 9.6%. But the flag catches only 21.0% of the busy outlooks. Fewer flags are borne out more often, but most busy outlooks get no flag.

Code decides, the agents explain, a person approves. Rules set every number and level, so they are repeatable and checkable. The language model adds explanation only, and every brief costs model calls and a person's time.

The model backend. The local model runs on this machine, costs nothing per call and is pinned by digest. It verified none in either round. In round 1 each retry went out blind, a fault in my code that calls the model, fixed before round 2, and the time budget stopped the run after 3 of the 7. In round 2 every draft reached the 8,192-token output cap, which counts the model's thinking; a retry with thinking turned off is untried and is a next step (section 14). The frontier model verified 5 of 7, short of the adequacy test's 6: the checks refused both demand-week briefs for claims their cited figures did not support, and the template writes those weeks in new runs (section 13). Its set is the one approved, at the cost of money, an outside service and a model that cannot be pinned the same way.

6. Constraints

1. **Rotas are set ahead.** Rosters reach resident doctors weeks before a rotation (NHS England, 2025), and rota guidance describes shifts known months in advance (RCEM, 2019). Neither source is Scottish. *Shaped*: the four release horizons and front-door staffing actions only.
2. **The publication lag.** Each Tuesday release covers the week that ended 9 days earlier. *Shaped*: the horizons count from the release.
3. **No beds, staffing or occupancy in the open data.** *Shaped*: the capacity is simulated and the levels illustrative.
4. **Demand is not pressure.** Attendances run highest in May and June, while 4-hour performance is worst in December and January (Figure 2). *Shaped*: the level is a demand level, never called pressure.
5. **Shocks such as COVID break every model.** *Shaped*: the drift latch, the surge rule and the re-selection at each latch reset.
6. **Escalation in Scotland.** The 2016 guidance sets three stages, triggered by hours after a patient is ready to move, department occupancy and ambulance delays (Scottish Government, 2016),

none of them attendances, so a forecast of attendances cannot set a stage. The October 2026 surge framework says "No single indicator should determine escalation decisions in isolation" (Scottish Government, 2026, Section 2). The Scottish Government committed to an NHS Scotland OPEL system in June 2025 (Scottish Government, 2025, recommendation 33). *Shaped*: the bands borrow only the structure of England's OPEL framework, and the level never sets an escalation stage.

7. **Runtime and tools.** `config.json`'s `llm_backend` is `frontier`: the frontier set is the one approved, and the recordings make the notebook reproducible on either setting. Setting it to `local` selects the local model. The local model did not write adequate briefs in either round, its retries blind in round 1 and its drafts cut at the output cap in round 2, so the frontier model ran once, as the adequacy test set out before the runs. It verified 5 of 7, short of the adequacy test's 6, as the checks refused both demand-week briefs, and the approver chose its set on those facts. The local model stays an option to retest with its thinking turned off. Each GPU run was meant to stay under 15 minutes. Round 1 stopped at its 840-second budget; for round 2 I allowed up to 60 minutes and set a 30-minute budget, and it used 657 seconds after a 95-second model load. The one technology not used in an earlier capstone project is the Anthropic client behind the frontier backend. It is a new library in an existing pattern, behind the same model gateway as Agentic AI Systems' OpenAI client, used only with the stated case the adequacy test set out before the runs. The notebook's replay needs neither it nor a key. *Shaped*: the bounded replay set, the local model and the backend switch.
8. **Time.** The aim was to submit within a week, with quality before the date. *Shaped*: a tight build, with shadow mode on new releases stated as a pre-deployment step.

7. Risks

#	Risk	Effect	Control	Owner
1	The front door is staffed for the usual week when demand is not usual (Scottish Government, 2016)	Harm from crowding; waits also depend on beds and flow, which the system does not see	Earlier notice of unusual demand beside the manager's usual information; the level's meaning on every brief	Duty manager
2	A shock breaks every forecast: the linear model's national forecast made at w/e 8 Mar 2020 missed by 152.8% three weeks later	Levels and briefs built on broken forecasts	The drift latch; the surge rule; forecasts marked untrusted; re-selection at each latch reset	Unscheduled-care lead (reset); analytics lead (re-selection)
3	False flags and alert fatigue, the habit of ignoring flags when many are false: the mean steady year raised 145 outlook flags, 66 of them not borne out. September's flags were borne out 40.4% of the time, against 14.9% for unflagged September outlooks	Staff time spent for nothing; briefs ignored	Approval of every brief; the hospital's track record on every brief; the sensitivity table; the flag reported by month	Unscheduled-care lead (thresholds)
4	A missed busy week: the flag caught 21.0% of busy outlooks in the steady years, and the mean steady year left 85.5 of 142.5 busy hospital-weeks with no notice	No extra notice of an unusual week, so staffing rests on the existing process alone	The range and the hospital's own error on every brief; recall and notice reported; learning what planners use (section 14)	Unscheduled-care lead

#	Risk	Effect	Control	Owner
5	Bias by hospital size: seasonal naive's median error is 11.0% at sites under 300 attendances a week and 5.4% at 1,300 and over. The raised share varies more by site than by size: the most raised, the Royal Hospital for Children at 35.7% of its outlooks, and Glasgow Royal Infirmary, raised in none, are both in the largest band, 1,300 attendances a week and over	Attention spread unevenly, by noise at small sites and not by need	Each brief shows its own site's error and track record; the level is never read as need or priority; the level reported by site	Unscheduled-care lead
6	Misuse as a league table, a case for cuts, or 4-hour monitoring	Harm to the hospitals the size bias already burdens	The level's meaning on every brief; the stated purpose in the README, the notebook and this case study	Unscheduled-care lead
7	Classification as a medical device (Medicines and Healthcare products Regulatory Agency, 2023)	Regulatory duties the design does not meet	Aggregate planning only; a code screen for the MHRA's indicative words; status as additional information; my reading kept as an open risk	Clinical safety reviewer
8	An invented or misattached figure	A manager misled by a wrong number or date beside their usual information	The code figure check on every brief, edit and notification; a "not verified" brief never approved; a planted figure caught in the test	Analytics lead (the check); duty manager (approval)
9	Automation bias (Goddard et al., 2012)	A manager approves without reading, or weighs a brief above their usual information	Ranges, disagreement and the track record shown; six approval criteria; a second reader in the evaluation	Duty manager
10	In-sample design tuning	Figures better than a new period would give	The notebook re-measures every figure; shadow mode on new releases before any go-live	Analytics lead
11	The flag's stated chance runs high: in the steady years each week's own chance averaged 0.162, and 8.9% of those weeks were busy	A manager reads a flag as surer than it is	The brief quotes the hospital's track record, never the chance	Analytics lead
12	The local model writes briefs too weak to approve	Nothing usable reaches approval, or weak briefs are approved	The adequacy test stated before any brief; one run of the frontier backend with a stated case; the Reviewer on either backend	Analytics lead

8. Boundaries of capability and responsibility

What the system does not do. At this stage it gives the duty manager additional information beside their existing process: it replaces no current process and sets no staffing on its own. It never sets a level by judgement, never sends without approval, never sees patient data and never quotes the flag's stated chance. It forecasts weeks, not days, and it does not see beds, flow or 4-hour

performance, so the level is never a measure of pressure. It is not for ranking hospitals, justifying cuts or monitoring the 4-hour standard.

Where a person takes over. The duty manager approves, edits or rejects every brief, and only an approved, verified brief leads to staffing actions, each assigned to a person who decides what to do. While the drift latch is set the forecasts are untrusted, and only a person approves a latch reset; in this build each reset is simulated at its review prompt. People acknowledge each action, and a missed deadline goes to the role above. The design follows the principle that a person can disregard, override or reverse the output (European Union, 2024, Article 14), as a principle, not a legal duty.

Owners by role.

Role	Owns
Duty manager	Each brief's decision, and the check on extra shifts and bank staff
Unscheduled-care lead	The thresholds and the bands; approval of each latch reset
Analytics lead	The re-selection of the primary model at each reset, and any other model change
Clinical safety reviewer	Sign-off before any live use

In the spot check I play the approver, the unscheduled-care lead and the analytics lead. The clinical safety reviewer's sign-off is not exercised, because there is no live use. Approving briefs for a system I built is a limitation, which a second reader, Reader B, checks: Reader B made the same decision as the approver on all four briefs.

An AI independent judge read the seven briefs before I decided. I adopted its five edits as my own, with the reset wording corrected, and its notes on the two rejections, as each decision's note records. An AI reviewer, not the system's Reviewer, scored the adequacy test, so my approval stayed separate from it.

9. Reuse register

Each pattern is adapted for this project, never copied in as a file, and each module names the pattern it reuses in a comment. Each repository is at github.com/PaddyGilliland1, and cell numbers count from 0.

Project	Repository	File: function or cell	Commit	Component here
AI Programming Foundations	ai-programming-foundations-project	`data_workflow.ipynb`: `standardise_columns` (cell 9), `flag_unexplained_payments` (cell 27)	`dc8dabd`	Load, clean, flag and keep (`data.py`)
Conduct a Statistical Analysis Using Python	statistical-analysis-project	`analysis.ipynb`: `mean_with_ci` (cell 47), the trend line with `linregress` (cell 101), `bootstrap_slope` (cell 104), `paired_t_test` (cell 122)	`7556623`	The flag's and the drift interval's spread of past errors, the linear model's trend, the moving-block bootstrap and the paired comparison (`rules.py`, `backtest.py`, `forecasts.py`)
Deep Learning Systems	deep-learning-systems-project	`deep_learning.ipynb`: `naive_forecasts` (cell 24), `ForecastRNN` with `cell="gru"` (cell 56), `predict` (cell 61)	`5511b2b`	The simple bars to beat and the GRU comparator (`forecasts.py`, `gru.py`)

Project	Repository	File: function or cell	Commit	Component here
Agentic AI Systems	agentic-ai-systems-project	`agentic_system.py`: `screen`, `Toolbox` with `verify_figures`, `ModelGateway` with `RecordingStore`, `approval_queue`, `AuditLog`, the case file and the manifest check	`d110982`	The screen, the typed read-only tools, the Investigator's tool loop, the Reviewer's figure check, record and replay, the approval queue, the audit log and the data manifest check (`tools.py`, `briefs.py`, `agents.py`, `figures.py`, `gateway.py`, `approval.py`, `replay.py`, `data.py`)

10. Method provenance

Method	Prior project, or new	Published source
Load, clean, flag and keep	AI Programming Foundations	None needed
Confidence intervals and a seeded bootstrap; the paired comparison of two models	Statistical Analysis	None needed
The linear model: a trend line plus four sine and cosine pairs	The trend line from Statistical Analysis; the sine and cosine terms my own extension	Hyndman and Athanasopoulos (2021), Section 7.4
The rolling-origin back-test, scored by the mean absolute percentage error (MAPE)	Statistical Analysis, its time-based hold-out repeated every week	Hyndman and Athanasopoulos (2021), Sections 5.10 and 5.8
The moving-block bootstrap	Statistical Analysis, its bootstrap resampled in blocks	Hyndman and Athanasopoulos (2021), Section 12.5; Künsch (1989)
Seasonal naive and the moving average as bars to beat	Deep Learning Systems	Hyndman and Athanasopoulos (2021), Section 5.2
SA-MA, the moving average with a seasonal index	New in this project	Described as my own extension
The probability flag	New, on Statistical Analysis's interval from past values	Described as my own design
The GRU comparator	Deep Learning Systems	Petravić et al. (2025) review recurrent networks among emergency department forecasters
Re-selection of the primary by out-of-sample error	Statistical Analysis	Hyndman and Athanasopoulos (2021), Section 5.10
The drift interval, the alarm and the latch	The interval from Statistical Analysis; the two-week alarm and the latch new	Hyndman and Athanasopoulos (2021), Section 5.5, for the interval's basis only
The screen	Agentic AI Systems	None needed
Agents with typed read-only tools, the approval queue, replay and the audit log	Agentic AI Systems	Yao et al. (2023) for tool use
The Reviewer: the code figure check, then the wording check	Agentic AI Systems' figure check, extended	Described as my own design
The frontier backend behind one configuration value	New, on Agentic AI Systems' model gateway	None needed

Built beyond the plan, new in this project:

- a code screen for the words the MHRA lists as indicative of a medical device;
- the tool-call cap, and a time budget that holds for a whole run;
- the "Written by" line naming the backend and model on every brief;
- the scripted demonstrations of the safeguards;
- the self-labelling placeholders: code writes every figure and what it is, and the model's words name no number, model, week, hospital or level;
- the Reviewer's third check, that each sentence describes its figures as their labels say;
- one more try for a Reviewer whose call failed or whose verdict could not be read;
- "not verified" counted by cause;
- the commands that record the adequacy test and the approver's choice of set.

11. Scenarios

The replay set is five named series, the national series and four hospitals, at six origins. The expected outcomes were written before any agent ran; the pre-run check found the same seven briefs.

Origin (release)	Expected briefs	Observed, frontier set
w/e 6 Oct 2019 (Tue 15 Oct 2019)	1: Royal Infirmary of Edinburgh, surge, its forecasts untrusted	Verified on the retry; approved with an edit. Its two notifications and one escalation reached the outbox, never sent
w/e 8 Dec 2019 (Tue 17 Dec 2019)	0: the week before that winter's worst national 4-hour week, with none of the five replay series flagged (the full screen sends 4)	No brief
w/e 15 Mar 2020 (Tue 24 Mar 2020)	0: the first breach, with no alarm yet among the five replay series (the full screen sends 1)	No brief
w/e 22 Mar 2020 (Tue 31 Mar 2020)	4: latch, for the national series, Edinburgh, Ninewells and the Queen Elizabeth; Lorn & Islands not latched	All four verified, one on the first attempt; each approved with an edit. Level 1, held, so no staffing action
w/e 13 Jun 2021 (Tue 22 Jun 2021)	1: Ninewells, demand, flagged	Not verified: the code check refused the first draft, and the wording check refused the second. It claimed a rising trend from one cited actual, though the history tool had returned the rising weeks, and repeated code's own line that the forecasts fit recent weeks. It reached the approver marked "not verified, do not act" and was rejected; in new runs the template writes this kind of week. The designed failure, a second run of this brief with a figure planted by code in its first readable draft, was caught at that figure
w/e 9 Apr 2023 (Tue 18 Apr 2023)	1: Lorn & Islands, demand, flagged	Not verified, as for Ninewells: its second draft repeated code's line that the forecasts fit recent weeks, with other claims its figures did not support. Rejected, with the same next step
Total	**7**: 4 latch, 1 surge, 2 demand	5 verified, approved with an edit; 2 rejected

Neither local set verified a brief, so none of its briefs could be approved. Round 1 completed 3 of its 7 briefs before its 840-second budget stopped it, and the text rules refused both drafts of each: each retry went out blind, a fault in my code that calls the model, fixed before round 2. Round 2 completed all seven, and every draft was cut at the 8,192-token output cap, which counts the model's thinking; a retry with thinking turned off is untried (section 14).

The template comparison used two of these weeks: the national latch week, w/e 22 Mar 2020, and Ninewells' demand week, w/e 13 Jun 2021. By option A, both raters found the agent's text added value in the national latch week and not in the demand week, whose brief the checks had refused. That result triggered the rule set before the ratings, so in new runs the template writes a flag week's explanation; section 13 gives the marks.

12. Plausibility notes

Each designed-only element, and what a live system would do instead.

- **The replay set.** Seven briefs from five series at six origins stand for the full screen, which at those origins would send 2, 4, 1, 12, 4 and 6 briefs across all 30 hospitals and the national series.
- **The GRU.** It runs on the five replay series here; in the design it would run per hospital, still as a comparator only.
- **Notifications.** They go to an outbox file and are never sent. A live system would send them through the board's own messaging, with the same check before each one.
- **Acknowledgements.** They are simulated and seeded. In use, people acknowledge each action.
- **Latch resets.** They are recorded at each review prompt. In use the unscheduled-care lead approves each one against documented reset criteria.
- **Capacity.** It is a demand percentile. A board would measure against its own staffed capacity.
- **The staffing actions.** Code fixes each action's text by level; the language model drafts only each notification's wording. A board would set its own actions and owners.
- **The roles.** I play three roles, and the clinical safety reviewer's sign-off is not exercised.
- **The template comparison.** Two raters marked two weeks, and one of them is the approver, whose marks are not blind. A pilot would repeat it with managers who did not build the system (section 14).
- **Shadow mode.** The frozen design has not yet run on new weekly releases.

13. Evaluation summary and limits

Goals G1 to G8, each against its threshold

Goal	Met if	Result	Verdict
G1 Forecasts a manager can plan with	within 1 point at every release horizon	-0.18 to -0.74 points against the best simple model	met
G2 Honest uncertainty	90% to 98%	95.7% to 96.4% of actuals inside	met
G3 Shocks are caught	by Tue 31 Mar 2020; at most 1 a year	known Tue 31 Mar 2020; at most 1 alarm a year	met
G4 No unchecked number approved	every approved brief passes; no not-verified brief approved; plant caught	5 of 5 approved pass; 2 not verified, neither approved; plant caught	met
G5 The agent never sets a level	always the rule's level	7 of 7 briefs written	holds
G6 Briefs are useful to a person	6 of the 7 briefs approved with no factual change, and Reader B makes the approver's decision on at least 3 of the 4 they read	5 of 7 (the two demand-week briefs were refused and rejected); Reader B agreed on 4 of 4	not met
G7 Raised levels rare in steady years	under 10% in every steady year	over 10% in 3 years, 10.4% to 11.7%	narrowly missed
G8 A flag tells a manager something	at least twice the base rate	2.54 times the base rate, steady years	met

G1 to G3, G7 and G8 are in-sample, on 2015 to 2026. G4 to G6 are on the frontier set's seven replay briefs. G5 is an invariant, not a test.

Figure 5 (notebook Figure 5). Goals G1 to G8, each against its threshold.

The goals. Five goals are met, the invariant holds, raised levels rare in steady years (G7) is narrowly missed in 2018, 2023 and 2024, and briefs useful to a person (G6) is not met because the checks refused both demand-week briefs.

- G1, forecasts a manager can plan with, G2, honest uncertainty, G3, shocks are caught, and G8, a flag tells a manager something, are met, in-sample. The primary's national error is within 1 point of the best simple model at every release horizon, the best simple model being the better of seasonal naive and the 4-week moving average at that horizon. Between 95.7% and 96.4% of national actuals fall inside its 95% range. The national drift alarm was known on Tue 31 Mar 2020. Outlook flags were borne out 2.54 times as often as the base rate in the steady years.

- G4, no unchecked number reaches a manager as approved, is met: all 5 approved briefs pass the code check again, neither brief that was not verified was approved, and the planted figure was caught. G5, the agent never sets a level, holds in all 7 briefs written.
- G6, briefs useful to a person, is not met: 5 of the 7 briefs were approved with no factual change, where 6 are needed; the two demand-week briefs were refused and rejected, since the approver could only reject a brief the checks refused. None of the five edits was factual. In new runs the template writes those weeks. Reader B made the approver's decision on all 4 of the 4 briefs they read, so G6's second half is met.
- G7, raised levels rare in steady years, is narrowly missed, as set before the run, and nothing was tuned: 10.4% of outlooks were raised in 2018 (from July), 10.9% in 2023 and 11.7% in 2024, against under 10%. My reading, which nothing here tests, is that in those years attendances ran above the three previous years that set the capacity. Shadow mode tests that reading on weeks the design has never seen, before any threshold moves (section 14).

The headline, in its two units. In the steady years, a flag means an unusually busy week is likely in the next four: that held 53.8% of the time, against 21.2% without a flag. The exact week named was right 27.6% of the time, against 7.8% for a week without a flag.

Notice, each hospital-week counted once. The mean steady year gave notice of 57.0 of 142.5 busy hospital-weeks (40.0%), 49.0 of them before the week began. This is the notice the flag alone would give, before any brief is verified or approved; in the replay set neither flag week's brief could be approved. Over the steady years, 280 of 713 busy hospital-weeks had notice, carrying 19,806 of 46,243 extra patients; 197 had it from the week's own flag.

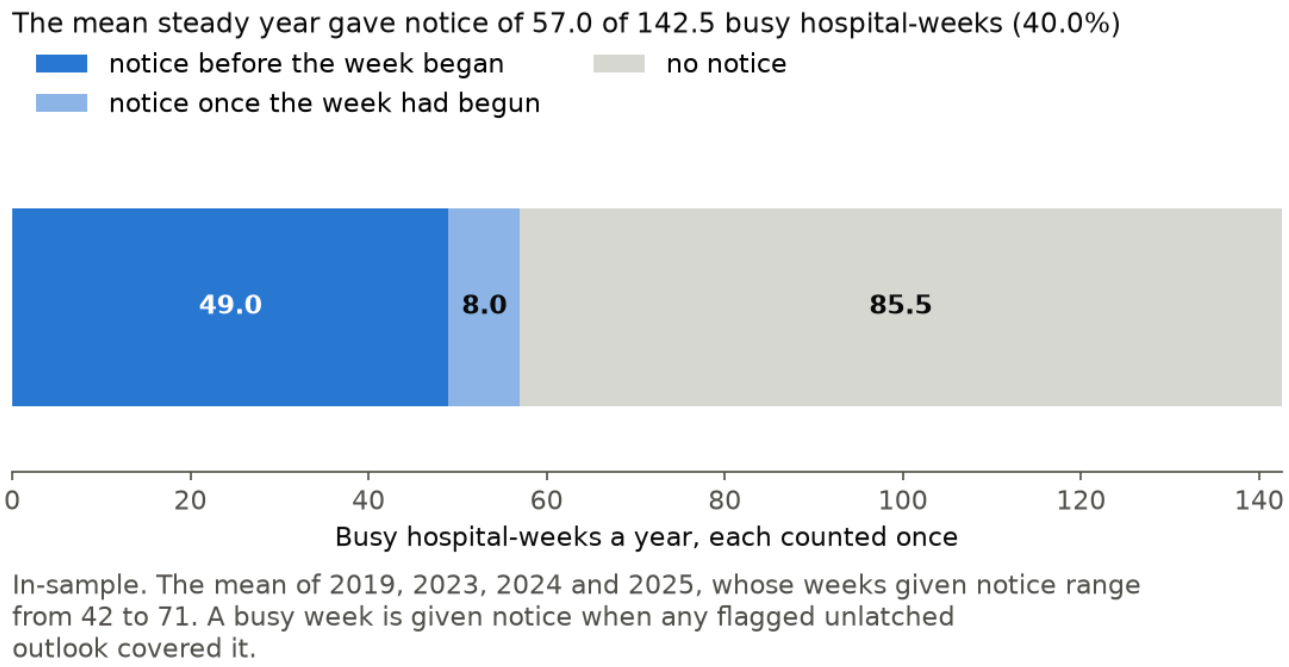


Figure 6 (notebook Figure 6). Busy hospital-weeks given notice, the mean steady year.

Calibration. Each week's own stated chance runs high: in the steady years it averaged 0.162, and 8.9% of those weeks were busy. So the brief quotes the hospital's track record, never the chance. The flag still beats each hospital's real-time busy share, with a Brier skill of 0.128. Why the chance runs high is not known; testing calibration by hospital and horizon comes before any threshold moves.

The September residue. 12.8% of steady-year September outlooks are flagged, and 40.4% of those flags are borne out, against 14.9% of unflagged September outlooks.

Demand is not pressure. In the steady years 29.9% of raised outlooks fell in their hospital's worst quarter of 4-hour weeks for the year, against 25.4% by chance: a weak link.

The template comparison. Both raters marked each text of both pairs against the six criteria, the two texts shown as Text A and Text B in an order hidden from Reader B; the approver had been told it. The approver's marks are not blind, since the approver had read every agent brief at approval. By option A, the agent's text adds value in the national latch week for both raters. Both counted it as naming the models' disagreement, which the template cannot, because it marked every model untrusted while latched and ranked none. In the demand week it does not, for both raters. That text is the refused demand-week brief. Both marked its figures not correct because it is marked not verified, though its figures had passed the code check. They also marked its uncertainty not stated, and a factual error in its claims that recent weeks were rising and the forecasts fit them. Both also marked a factual error in the template's drift line, the line code gave the Investigator too. The shipped template, version 2, and new runs now use the raters' own wording, "No drift alarm is recorded; this alone does not establish forecast fit.", and the comparison stays on the rated version 1.

The comparison failed in the demand week, whose brief the checks had already stopped, so no manager could approve it. A rule set before any rating then applies. The template writes the brief's explanation by default, and the agent only in drift, surge or model-disagreement weeks. `config.json`'s `explanation` setting now applies it in new live runs: a flag week gets the template brief, filled by code and queued for approval like any brief, and a latched week the agent, as section 3 sets out. The recorded runs had the agent write every brief and replay as they ran. The rule's model-disagreement ground has no threshold set before the ratings, so no week is kept for the agent on that ground; defining one is a next step. Neither pair was blind to Reader B, since each agent text was one of the four briefs Reader B had rated, and the template was Text A in both weeks.

Template against agent: both raters, both weeks

	National latch week w/e 22 Mar 2020				Demand week w/e 13 Jun 2021			
	The approver		Reader B		The approver		Reader B	
	template	agent	template	agent	template	agent	template	agent
Figures correct	●	●	●	●	●	○	●	○
Uncertainty stated	●	●	●	●	●	○	●	○
Model disagreement named	○	●	○	●	○	●	○	●
The level's meaning stated	●	●	●	●	●	●	●	●
Watch points actionable	●	●	●	●	●	●	●	●
No clinical instruction	●	●	●	●	●	●	●	●
Factual error	no	no	no	no	yes	yes	yes	yes
Option A: agent adds value	yes		yes		no		no	

Filled: the rater marked the criterion met; open: not met. Option A counts a factual error in the agent's text only. Neither pair was blind to Reader B: each agent text was one of the four briefs Reader B had rated, and the demand week's carried its NOT VERIFIED banner. The template was shown first, as Text A, in both weeks. The approver's marks are not blind.

Figure 7 (notebook Figure 7). Template against agent: both raters, both weeks.

Failure cases. COVID, real: the alarm confirms a shock and does not warn; the first breach was w/e 15 Mar 2020. The latch then held each series' level and marked its forecasts untrusted, and shadow

mode on new releases is the next step. The planted figure, designed: caught at that figure. The demand-week briefs, real: the Reviewer refused Ninewells' rising trend that cited one actual and, in both, code's own line that the forecasts fit recent weeks, so neither could be approved, and in new runs the template writes those weeks. The local backend: no brief verified in either round. Round 1's retries went out blind, a fault in my code that calls the model, fixed before round 2, and its budget stopped it after 3 of 7; round 2's drafts all reached the output cap, and thinking turned off is the untried next step.

Limits. The notebook's section 9 sets out seventeen limitations, each with its evidence, what held it and the next step. In short:

1. the track-record line drops the unflagged busy share when a hospital has no past flag; the next step states it;
2. a back-test error sits beside "forecasts untrusted"; the approved edits say it is historical, and code will say so next;
3. the Reviewer's reasons are noisy, though its verdicts held; the next step asks it for failures only;
4. the local model was not tried with its thinking turned off; the checks let no draft through, and that trial is next;
5. explaining a demand week overreached; the checks stopped both briefs, in new runs the template writes those weeks, and a trend must cite the weeks it rests on;
6. the rated drift line claimed the forecasts fit recent weeks on the drift rule alone; the shipped template and new runs now use the raters' wording, and the comparison stays on the rated version;
7. neither template pair is blind to Reader B; a blind comparison on new weeks is the next step;
8. the template comes first in both pairs; a new comparison draws the order for each week;
9. a level-2 action holds leave while the forecasts are untrusted; the approved edits removed two blanket holds, and the next step makes the hold a separate decision;
10. a range can join the ends of two weeks; the approver's note caught it, and a code check that both ends share a week is next;
11. G6 is not met, as both demand-week briefs were refused and rejected, with the builder as approver; Reader B made the same decisions;
12. round 1's retries went out blind, a fault in my code that calls the model, fixed before round 2;
13. a notification's wording is checked for its figures and words, not its meaning; a wording check on each notification is next;
14. the Reviewer reads the brief inline, so a draft could address it; the code check runs first, and marking the brief off from the instructions waits for a new run;
15. G1 and G2 are scored on a mask set afterwards; scoring the forecasts trusted when issued is next;
16. the back-test reads one revised snapshot; archiving each release, or measuring the revisions, is next;
17. the flag's stated chance runs high, and the cause is not known; testing calibration by hospital and horizon comes before any threshold moves.

Every figure here is in-sample.

14. Deployment path and next steps

At this stage the system gives additional information beside the duty manager's existing process, by choice. These are future pre-deployment investigations after this base is proven, in order. Together they are the path from that additional information to a primary operational tool.

1. **Shadow mode before any go-live.** The frozen design runs on new weekly releases, every brief written and logged and none acted on, and the goals are scored on weeks the design has never seen. Before it starts, two gaps this build found need closing: a trend claim that cites one week,

and a trial of the local model with its thinking turned off. A third, the drift line's claim that the forecasts fit recent weeks, is closed in the shipped template and in new runs. The template rule's model-disagreement ground also needs a threshold.

2. **A pilot that measures throughput**, link 5 of the value chain, with managers who did not build the system, and bands set per hospital for fairness. Within the pilot, the managers' feedback includes learning what planners use.

   Learn what planners use. Ask the duty managers what they weigh beyond the tool, and measure which factors would have explained past busy weeks. Then route each one:

   - a model input, if it recurs with a measured history, tested on the back-test;
   - a manager parameter, if it is known in advance but one-off: a labelled, recorded adjustment, like an authorised forecast adjustment in enterprise resource planning (ERP) demand planning;
   - human judgement at the approval step, if it cannot be measured.

   The named candidates, as illustrations only, are weather, population demographics, and flu strains with their expected severity.

   Tuning over time then acts on what shadow mode and the pilot find, the managers' feedback and what planners use among it.

3. **The wider system.** Once the base is proven, admissions and occupancy data widen it, so the same forecast, flag, explain, approve and act pattern can speak to beds, overflow wards and earlier contact with social care.

Only after shadow mode, the pilot and tuning over time could the system move from additional information to a primary operational tool, and the wider system builds on that proven base.

References

Audit Scotland. (2025, December 4). *NHS in Scotland 2025: Finance and performance*. https://audit.scot/uploads/2025-11/nr_251204_NHS_in_Scotland_2025.pdf

European Union. (2024). Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act), as amended by Regulation (EU) 2026/1744. *Official Journal of the European Union*, L, 2024/1689. http://data.europa.eu/eli/reg/2024/1689/oj

Goddard, K., Roudsari, A., & Wyatt, J. C. (2012). Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association, 19*(1), 121-127. https://doi.org/10.1136/amiajnl-2011-000089

Hu, Y., Chan, C. W., Dong, J., Kazekjian, A., Ophaswongse, C., Sugalski, G., Underwood, J. P., & Perotte, R. (2025). Implementing a prediction driven framework for emergency department nurse staffing to optimize real time decisions. *npj Health Systems, 2*(1), Article 16. https://doi.org/10.1038/s44401-025-00019-2

Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. https://otexts.com/fpp3/

Imamura, Y., Sakuramoto, H., Ouchi, A., Honda, T., Uno, S., & Tamoto, M. (2026). Emergency nurse staffing and outcomes for patients and nurses: A scoping review. *Journal of Emergency Nursing*. Advance online publication. https://doi.org/10.1016/j.jen.2026.05.015

Jones, S., Moulton, C., Swift, S., Molyneux, P., Black, S., Mason, N., Oakley, R., & Mann, C. (2022). Association between delays to patient admission from the emergency department and all-cause 30-day mortality. *Emergency Medicine Journal, 39*(3), 168-173. https://doi.org/10.1136/emermed-2021-211572

Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics, 17*(3), 1217-1241. https://doi.org/10.1214/aos/1176347265

Medicines and Healthcare products Regulatory Agency. (2023). *Medical device stand-alone software including apps (including IVDMDs)* (Version 1.10g; file revised October 2026). GOV.UK. Retrieved October 24, 2026, from https://assets.publishing.service.gov.uk/media/6abf6c8c943e890cb55e5e52/Medical_device_stand-alone_software_including_apps__including_IVDMDs.pdf

NHS England. (2024, December 2). *Integrated operational pressures escalation levels (OPEL) framework 2024 to 2026* (page updated 23 May 2025). https://www.england.nhs.uk/publication/integrated-opel-framework-2024-to-2026/

NHS England. (2025). *Resident Doctors (RDs): Rota Code of Practice compliance data collection* (DCB3042 AMD 35/2025) [Data collection]. https://standards.nhs.uk/published-standards/resident-doctors-rds-rota-code-of-practice-compliance-data-collection

Petravić, L., Gril Rogina, K., Albreht, T., Kukec, A., & Žibert, J. (2025). Prognostic models for predicting patient arrivals in emergency departments: An updated systematic review and research agenda. *BMC Emergency Medicine, 25*(1), Article 106. https://doi.org/10.1186/s12873-025-01250-8

Public Health Scotland. (2026a). *Weekly A&E activity and waiting times* [Data set]. Scottish Health and Social Care Open Data. Retrieved October 24, 2026, from https://www.opendata.nhs.scot/dataset/weekly-accident-and-emergency-activity-and-waiting-times

Public Health Scotland. (2026b, October 6). *Accident and emergency: Main points, all A&E sites*. https://publichealthscotland.scot/healthcare-system/urgent-and-unscheduled-care/accident-and-emergency/main-points/all-ae-sites/

Public Services Delivery Scotland. (2026, June 2). *NHSScotland workforce: Latest statistics at 31 March 2026* [Accredited official statistics]. Turas Data Intelligence. https://turasdata.nes.nhs.scot/data-and-reports/official-workforce-statistics/all-official-statistics-publications/02-june-2026-workforce/?pageid=16251

Rivet, C., Amazouz, H., Gratacap, M., Delannoy, Q., Yordanov, Y., Leblanc, J., & URGE collaborators. (2026). Association between patient-to-nurse ratios in emergency departments and patient, staff, and organizational outcomes: A systematic review. *European Journal of Emergency Medicine, 33*(4), 237-247. https://doi.org/10.1097/MEJ.0000000000001350

Royal College of Emergency Medicine. (2019). *EM-POWER: A practical guide to flexible working and good EM rota design*. https://rcem.ac.uk/wp-content/uploads/2021/10/EM-POWER_A_practical_guide_to_flexible_working_and_good_EM_rota_design.pdf

Royal College of Emergency Medicine. (2026). *The state of emergency medicine in Scotland* [Report]. https://rcem.ac.uk/wp-content/uploads/2026/03/RCEM_SoEM_Scotland_A4_0326.pdf

Scottish Government. (2016). *Emergency department capacity management: Guidance*. https://www.gov.scot/publications/emergency-department-capacity-management-guidance-2/

Scottish Government. (2025, June 30). *NHS Greater Glasgow and Clyde emergency department review: Response*. https://www.gov.scot/publications/nhs-greater-glasgow-and-clyde-emergency-department-review-response/pages/response/

Scottish Government. (2026, October 8). *Health and social care - surge preparedness: National planning priorities and principles*. https://www.gov.scot/publications/health-social-care-surge-preparedness-national-planning-priorities-principles/

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. In *The Eleventh International Conference on Learning Representations (ICLR 2023)*. https://openreview.net/forum?id=WE_vluYUL-X